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1 D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-
  PIKAN_Official\experiments\Volterra1D_MCPI.py
2 The network model is PINNs
3 Layers: ModuleList(
4   (0): Linear(in_features=1, out_features=20, bias=True)
5   (1): Linear(in_features=20, out_features=20, bias=True)
6   (2): Linear(in_features=20, out_features=1, bias=True)
7 )
8 Activation function: Tanh()
9 Total trainable parameters: 481
10 Model on device: cuda:0
11 Start training...
12 Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
13 Training: 100%|██████████| 2000/2000 [00:06<00:00, 329.97epoch/s, Loss=1.
  64e-04, LR=1.0e-03, Best=1.64e-04]
14 Training finished. Best loss: 9.94e-05. Model saved to results/best_model.pth
15 Loss history saved to results/loss_history.npy
16 Training time: 6.934643268585205s
17 Relative loss: 0.011906294152140617
18 L2 loss: 0.007575991563498974
19 Parameter containing:
20 tensor([[ -1.2675],
21         [ -0.1430],
22         [  0.8256],
23         [-1.1236],
24         [-1.7957],
25         [  0.3312],
26         [-0.5673],
27         [  1.5062],
28         [  0.9710],
29         [-0.5409],
30         [-1.2744],
31         [-0.6004],
32         [  1.6032],
33         [-0.0783],
34         [  0.0393],
35         [-0.5987],
36         [-0.4224],
37         [-0.3118],
38         [  1.3215],
39         [-0.7316]], device='cuda:0', requires_grad=True)
40 Parameter containing:
41 tensor([ 1.0953, -0.1511, -0.2569,  0.8297,  1.5755, -0.4151, -0.1288,  0.2078,
42         -0.1960, -0.5397, -0.6711, -0.2055, -0.4125,  0.6651,  0.9365, -0.3826,
43         -0.1991, -0.4855,  0.2100, -0.6932], device='cuda:0',
44         requires_grad=True)
45 Parameter containing:
46 tensor([[ 3.6304e-01, -1.9878e-01,  2.0389e-02,  2.9949e-01,  3.4259e-01,
47         6.9305e-02,  1.9521e-01, -1.1226e-02, -1.4340e-01,  1.3458e-01,
48         3.2399e-02,  2.2982e-02, -5.8999e-02,  2.1690e-01,  1.1149e-01,
49         -9.8365e-03, -1.2422e-01,  4.8991e-02, -1.2529e-01,  2.0814e-02],
50         [-9.0207e-01, -1.4996e-01,  8.3394e-01, -1.2725e+00, -1.1370e+00,
51         2.2986e-01, -1.9652e-01, -4.3169e-02,  4.6786e-01,  2.3008e-01,

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52	1.5423e-01, -2.5717e-01, -1.7529e-01, -2.0264e-01, -2.0582e-01,
53	1.9143e-02, -1.9107e-01, -1.3582e-01, -2.1954e-01, 1.1500e-01],
54	[2.6388e-01, -9.9278e-02, -2.2051e-01, 3.5225e-01, -3.2234e-02,
55	9.9076e-02, 3.1828e-02, 2.0571e-01, -6.6034e-02, 9.0762e-02,
56	9.8712e-02, 1.1652e-01, -1.0276e-01, 5.6568e-02, 1.1434e-01,
57	-1.9382e-01, 1.0132e-01, 5.8566e-02, -1.4199e-02, -2.7788e-02],
58	[-6.8424e-03, -2.7460e-01, 2.8865e-02, -3.2101e-01, -2.0469e-01,
59	-1.4389e-01, -1.3591e-01, 1.6285e-01, 1.6623e-01, 1.1557e-01,
60	1.3922e-01, -2.0002e-01, -2.3465e-01, 4.3720e-02, -1.2881e-01,
61	1.5875e-01, 1.0985e-01, -7.8847e-04, 6.4255e-02, -1.3840e-01],
62	[1.7231e-01, 2.3643e-02, -5.0017e-01, 5.4697e-01, 4.3165e-01,
63	-2.6376e-01, 2.1812e-01, -8.8390e-02, -4.7344e-01, 1.3169e-01,
64	1.8522e-01, 2.4418e-01, -1.8440e-01, 9.9919e-03, -5.4327e-02,
65	1.9238e-01, -8.1580e-02, -1.3629e-01, 9.8749e-02, 3.9419e-03],
66	[8.5381e-02, 5.8508e-02, -9.5875e-02, -1.7095e-02, 9.7943e-02,
67	1.5655e-02, -1.0252e-02, -8.6096e-02, -2.4576e-01, 8.3621e-02,
68	-4.3655e-02, -1.5522e-01, 1.6140e-01, 7.9051e-02, -2.1833e-02,
69	-9.0305e-02, -7.9494e-03, 1.6766e-01, -8.4486e-02, 1.3736e-01],
70	[-7.4806e-02, 1.1987e-01, -1.2375e-01, -8.1775e-02, -3.4094e-01,
71	1.9625e-01, 5.8563e-02, 1.6457e-01, 5.1512e-02, 1.9152e-01,
72	2.0032e-01, -6.5508e-02, 1.4542e-01, 9.5401e-02, -1.3344e-01,
73	7.2087e-02, 1.5137e-02, 7.9930e-02, 5.3870e-02, -1.1704e-02],
74	[-9.1726e-02, 1.7022e-01, -8.4874e-02, 1.3940e-01, 3.0684e-01,
75	2.4575e-01, 7.4567e-02, -1.8447e-03, -1.1348e-02, 2.1651e-01,
76	1.3254e-01, 3.8965e-02, 2.7402e-01, -3.7266e-02, 1.4526e-01,
77	-1.1011e-01, 1.1935e-02, 2.7403e-01, 2.1731e-01, 8.9030e-02],
78	[-8.7004e-02, -1.4820e-01, -7.9623e-02, 5.7960e-02, -2.4188e-01,
79	6.0202e-02, 1.4369e-01, 1.3985e-01, -9.0108e-02, -2.2083e-01,
80	-4.3586e-02, -1.1291e-02, -1.7717e-01, 2.2116e-01, -5.2458e-02,
81	-2.5066e-02, -1.7557e-01, -2.1356e-02, -1.2843e-01, -1.4954e-01],
82	[3.0681e-01, 9.8700e-02, -6.6684e-01, 2.2884e-01, 3.2197e-01,
83	-9.0545e-02, 1.0959e-01, 1.3539e-03, -2.3245e-01, 1.5624e-01,
84	8.7443e-02, 1.6402e-01, -3.6924e-01, -1.5221e-01, -8.3889e-02,
85	1.7965e-01, 8.4243e-02, 2.1755e-01, 7.2460e-02, 5.8416e-02],
86	[3.5343e-01, 1.3747e-01, -2.3824e-01, 7.5478e-01, 4.9297e-01,
87	-3.6939e-01, 1.0895e-01, 2.3245e-01, -8.5415e-02, 5.4768e-02,
88	-1.5275e-01, 7.7011e-02, -1.3516e-01, 1.0300e-01, 3.4160e-03,
89	9.0193e-02, 2.0625e-01, -5.3196e-02, 9.2909e-02, -4.7201e-02],
90	[1.8496e-02, -2.4737e-02, -1.9338e-02, -6.6273e-02, -7.4673e-02,
91	-7.4187e-02, -1.0440e-01, 1.3756e-01, -2.4610e-02, 2.2564e-01,
92	1.3369e-01, 7.6945e-02, 1.5653e-02, 7.4250e-02, -3.5190e-02,
93	1.0861e-01, -6.5406e-02, 1.3457e-03, -1.3863e-01, -1.7645e-02],
94	[-9.6005e-02, -9.0359e-02, -1.5350e-01, 6.3988e-02, 9.2176e-02,
95	-1.9136e-02, 1.2062e-01, 3.4623e-02, -1.0798e-01, 9.6171e-02,
96	-6.5134e-02, 5.6216e-02, 1.4814e-01, 1.3035e-02, 1.5690e-01,
97	4.9326e-02, 2.6189e-01, -7.0654e-02, 1.7488e-01, -3.8729e-02],
98	[6.4780e-02, -2.6438e-01, 3.0089e-02, 1.1845e-02, 7.9093e-02,
99	-2.1678e-01, -2.3929e-01, -1.6480e-01, 6.5998e-02, -7.9680e-03,
100	-2.0718e-01, -5.0288e-02, 5.0171e-02, -1.4597e-01, -1.0858e-01,
101	1.5473e-01, -1.0919e-01, -1.7219e-01, -3.0981e-02, -2.1964e-01],
102	[-4.4814e-01, -3.8116e-01, 6.6391e-01, -5.2822e-01, -4.5383e-01,
103	3.8993e-01, -6.0056e-01, 4.9952e-01, 8.3066e-01, -4.6019e-01,
104	-2.6383e-01, -3.3365e-04, 5.5999e-01, 2.2313e-01, 3.2526e-02,

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105     -3.6731e-01, -2.9012e-01, -8.4265e-02,  5.2156e-01, -3.4064e-01],
106     [-3.7417e-02, -7.6590e-02,  1.5773e-01, -2.0511e-01, -7.5707e-04,
107     2.9295e-01, -3.0738e-02,  2.4275e-01,  3.3147e-01,  2.7366e-02,
108     -9.8169e-02, -1.4546e-01,  1.9162e-01, -1.2922e-01, -1.9766e-01,
109     -6.2214e-02,  1.9362e-01, -1.0391e-02, -9.7079e-03, -2.1670e-01],
110     [ 9.3720e-02, -1.3028e-01, -1.7958e-01, -2.1760e-01, -8.8646e-02,
111     -4.5906e-02,  1.5866e-01, -1.4866e-01,  1.2616e-02, -2.4526e-01,
112     -7.7443e-02, -2.5390e-01,  3.9869e-02,  1.9021e-01,  2.6487e-01,
113     1.1188e-01,  9.3018e-02, -6.9664e-03, -4.1605e-02,  2.4383e-02],
114     [ 5.6023e-02, -9.0776e-02, -5.0973e-01,  4.1554e-01,  3.2515e-01,
115     -2.3065e-01,  1.6824e-01, -1.5297e-01, -1.2454e-01,  1.7164e-01,
116     1.5914e-02,  1.3702e-01, -2.4297e-01,  4.2242e-02,  4.4160e-02,
117     -7.4626e-02,  1.7111e-01, -2.0839e-01, -8.7747e-02,  2.0005e-01],
118     [-1.6557e-01,  3.0018e-02, -7.3597e-02, -1.0989e-01, -1.2384e-01,
119     1.7689e-02, -1.5002e-01,  7.0633e-02,  1.4454e-01, -1.9231e-01,
120     1.1483e-01,  4.8137e-02, -2.7519e-01,  3.6227e-02, -7.8152e-02,
121     9.4501e-02, -2.3750e-01, -2.0881e-01, -1.3236e-01, -1.7326e-01],
122     [ 2.7659e-01, -9.1822e-03, -2.0579e-02,  6.1290e-01,  1.0778e-01,
123     -2.4960e-01,  1.4136e-01, -7.7575e-02, -1.9482e-01, -1.6446e-02,
124     -1.8269e-01,  6.8004e-02, -2.1051e-01,  2.1833e-01,  1.7522e-01,
125     2.2460e-01, -5.4985e-02, -1.6580e-01, -6.7857e-04,  5.7342e-02]],
126     device='cuda:0', requires_grad=True)
127 Parameter containing:
128 tensor([ 0.2428, -0.2240, -0.0352, -0.0792, -0.1712,  0.1107,  0.1653, -0.0190,
129         0.1851,  0.1806,  0.1427, -0.2209, -0.2346,  0.1864, -0.1742, -0.0580,
130         0.0623, -0.1542,  0.0771,  0.2218], device='cuda:0',
131         requires_grad=True)
132 Parameter containing:
133 tensor([[ 0.2557, -0.1417,  0.1508, -0.1474, -0.1429,  0.0052, -0.0880,  0.2946,
134         -0.1648, -0.2035,  0.4597,  0.1059,  0.0564, -0.1890,  0.4213,  0.2358,
135         -0.1927, -0.1048, -0.3823,  0.1308]], device='cuda:0',
136         requires_grad=True)
137 Parameter containing:
138 tensor([-0.0847], device='cuda:0', requires_grad=True)
139 =====
140 Layer (type:depth-idx)          Output Shape          Param #
141 =====
142 PINNModel                      [1, 1]                --
143 |—ModuleList: 1-5              --                      (recursive)
144 |   |—0.weight                  |—20
145 |   |—0.bias                    |—20
146 |   |—1.weight                  |—400
147 |   |—1.bias                    |—20
148 |   |—2.weight                  |—20
149 |   |—2.bias                    |—1
150 |   |—Linear: 2-1              [1, 20]              40
151 |   |   |—weight                |—20
152 |   |   |—bias                  |—20
153 |—Tanh: 1-2                    [1, 20]              --
154 |—ModuleList: 1-5              --                      (recursive)
155 |   |—0.weight                  |—20

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156 |      | 0.bias | 20
157 |      | 1.weight | 400
158 |      | 1.bias | 20
159 |      | 2.weight | 20
160 |      | 2.bias | 1
161 |      | Linear: 2-2 | [1, 20] | 420
162 |      | | weight | 400
163 |      | | bias | 20
164 | Tanh: 1-4 | [1, 20] | --
165 | ModuleList: 1-5 | -- | (recursive)
166 |      | 0.weight | 20
167 |      | 0.bias | 20
168 |      | 1.weight | 400
169 |      | 1.bias | 20
170 |      | 2.weight | 20
171 |      | 2.bias | 1
172 |      | Linear: 2-3 | [1, 1] | 21
173 |      | | weight | 20
174 |      | | bias | 1
175 | =====
176 |      |
177 | Total params: 481
178 | Trainable params: 481
179 | Non-trainable params: 0
180 | Total mult-adds (Units.MEGABYTES): 0.00
181 | =====
182 |      |
183 | Input size (MB): 0.00
184 | Forward/backward pass size (MB): 0.00
185 | Params size (MB): 0.00
186 | Estimated Total Size (MB): 0.00
187 | =====
188 |      |
189 | 进程已结束, 退出代码为 0

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